Smart Waste Segregation Assistant

A Real-Time Eco-Friendly AI System for Multi-Class Waste Classification

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***Abstract*—Improper waste segregation is still an environmental problem because of inconsistent recycling practices in various states in the United States. This paper discusses the development of a Smart Waste Segregation Assistant using computer vision and transfer learning technology for waste classification in real-time into five categories, including glass, metal, organic, paper, and plastic. The proposed approach uses the pre-trained MobileNetV2 model on ImageNet dataset and fine-tuned the classifier on TrashNet and RealWaste datasets with total 4,992 training images. Two-step fine-tuning of the classification head with the pretrained model followed by complete fine-tuning gives an accuracy of 90.97% with F1-score of 0.905 on held-out test set. The proposed real-time interface is implemented using OpenCV which shows predicted class name, confidence level, and disposal information. An ablation study was conducted to evaluate the effectiveness of adding glass as an additional class compared to four classes with accuracy loss of only 1.75%.**

***Index Terms*—waste classification, transfer learning, convolutional neural networks, real-time inference, MobileNetV2**

I. Introduction

Waste management is one of the greatest environmental concerns facing humanity today. The World Bank estimates that the amount of waste produced globally will increase to 3.4 billion tons by 2050 [1]. In America, different recycling policies adopted in each state cause confusion among its citizens, who often mis-sort their recyclables, ending up in landfills. This issue can be narrowed down to a single point of identification: the average individual does not know how to identify which waste item is made of glass, plastic, metal, paper, and organic material.

Computer vision offers a practical solution to this problem. A camera-based system can observe a piece of waste, classify it in real time, and immediately instruct the user where to dispose of it. This kind of point-of-disposal guidance has the potential to substantially reduce recycling contamination without requiring users to memorize complex sorting rules.

This project develops a Smart Waste Segregation Assistant that performs real-time five-class waste classification via webcam. The system uses MobileNetV2 as a lightweight CNN backbone, chosen for its efficiency on CPU-only hardware, and fine-tunes it on a combined dataset drawn from two publicly available sources. The interface provides category prediction, confidence percentage, color-coded feedback, and a plain-language disposal recommendation for each frame. An optional voice output feature is also included.

II. Related Work

There have been many efforts in recent times to classify trash through deep learning. In this context, the TrashNet dataset [6], which was proposed by Yang and Thung, provided the early benchmark with 2,527 images in six classes and became the most popular database for this problem.

Transfer learning for waste classification. The performance of various CNNs for recyclable waste classification based on the TrashNet database was investigated by Ahmed et al. [2]. These authors compared four CNNs, including the baseline CNN, DenseNet169, MobileNetV2, and ResNet50V2. Transfer learning models showed much higher efficiency than those that were trained from scratch; the accuracy was 98.95% for ResNet50V2 and 97.60% for MobileNetV2.

Real-time classification systems. Gondal et al. [3] propose a real-time waste classification model for smart cities that uses a hybrid architectural design consisting of a multi-layered perceptron for binary classifying metal vs. non-metal and then uses CNN for further classifying non-metal wastes. This design achieves over 90% classification accuracy and shows that there is a possibility of designing camera-based waste sorting models.

Comparison between architectures using waste datasets. Deng et al. [4] compared CNNs and Vision Transformers on TrashNet and TACO datasets. Though Vision Transformers can compete with CNNs in terms of performance, they consume many more resources (more training data and higher computing cost). Therefore, it supports the usage of CNNs in our research due to their lightweight property.

MobileNetV2 architecture. MobileNetV2, presented by Sandler et al. [5] at CVPR 2018, uses depthwise separable convolution blocks which help design a lightweight network named 'inverted residual block.' With approximately 2.55M parameters, MobileNetV2 architecture is much smaller than ResNet50 (25M) and DenseNet169 (14M).

III. Methods

A. Dataset and Class Mapping

The system used a mixture of two public datasets for training. TrashNet [6] has 2,527 images classified into six categories: cardboard, glass, metal, paper, plastic, and trash. The RealWaste dataset [7] consists of images taken under

real-life scenarios in nine different categories. The datasets were mapped to a five-category class scheme where cardboard is grouped together with paper, trash and food organics are grouped as organic, while textile trash, miscellaneous trash, and vegetation from the RealWaste dataset were excluded. The dataset was partitioned as follows: 70% for training, 15% for validation, and 15% for testing.

The final training set had 4,992 images where paper had 1,370 images, plastic 982, metal 840, glass 900, and organic 900 images. The images for organic and glass were augmented to 900 images each using offline data augmentation techniques such as flipping, rotations, color jittering, brightness/contrast changes, and cropping, which were then stored to disk prior to training.

B. Model Architecture

The base architecture is MobileNetV2 [5], pre-trained on ImageNet. The original classifier head is replaced with a custom head consisting of Dropout($p = 0.3$), Linear(1280, 256) with ReLU activation, Dropout($p = 0.2$), and a final Linear(256, 5) output layer. The total parameter count is approximately 2.55M, of which 329,221 are trainable during Phase 1 when the backbone is frozen.

C. Training Procedure

The training was performed using two-step procedure. During step one the MobileNetV2 was freezed and only classification head was trained for 30 epochs with Adam optimizer (learning rate= 1×10^{-3}, weight decay= 1×10^{-4}) with StepLR scheduler (steps = 10, gamma= 0.1). On step two, all layers were un-freezed and trained together with learning rate = 1×10^{-4} and StepLR (steps= 5, gamma= 0.1) for 15 epochs. The batch size was 32 and cross-entropy loss was used.

For online augmentations during training, we used random crop (from 256 to 224), flips, rotations up to 30°, color jitters, random affine transform, random perspective transforms, grayscaling, and random erasing. For validation and test images, resizing to 224×224 was applied and normalization.

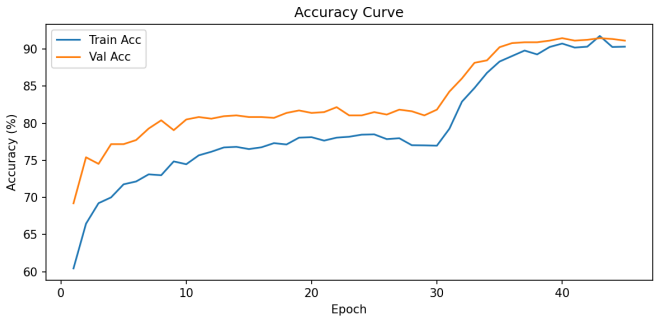
D. Real-Time Interface

In the real-time detection process, we make use of the OpenCV library to capture the webcam frames. These frames are then used in the inference transformation and forward processes with the pre-trained model. The output is annotated on top of each frame in form of the predicted label, softmax probability, recommendation for disposal, and the color bar showing the softmax probability. In case of a softmax probability of less than 50%, orange will show up.

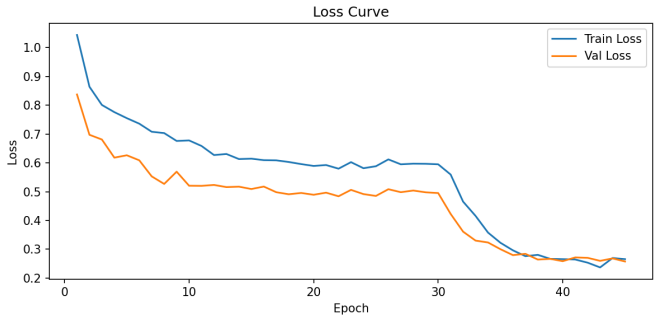
IV. Experiments and Results

A. Training Curves

The graphs for the accuracies and losses of the five classes through all 45 epochs for both training and validation can be observed in Fig. 1a and Fig. 1b respectively. The sudden spike in accuracy at epoch 31 is the distinction between Phase 1 and Phase 2. Overfitting does not occur in Phase 2.



(a) 5-Class model training and validation accuracy over 45 epochs.



(b) 5-Class model training and validation loss over 45 epochs.

Fig. 1: 5-Class model training curves.

B. Five-Class Model Performance

The five-class classifier yielded the best validation accuracy of 91.47% and test accuracy of 90.97%. Table I reports per-class precision, recall, and F1-score on the 908 images in the test dataset. The macro-average F1-score is 0.905.

TABLE I: Per-Class Results, 5-Class Test Set (n=908)

Class	Precision	Recall	F1
Glass	0.933	0.899	0.916
Metal	0.799	0.972	0.877
Organic	0.905	0.916	0.910
Paper	0.966	0.956	0.961
Plastic	0.939	0.796	0.862
Macro Avg	0.908	0.908	0.905

The paper presented the maximum F1-score of 0.961. Metal has the maximum recall of 0.972, but it has the minimum precision of 0.799. Plastic recorded the lowest F1-score of 0.862 and recall of 0.796. This result may be attributed to the visual similarity between glass and plastic containers, as shown in Fig. 2.

C. Four-Class Ablation Study

For assessing the impact of having glass as a distinct class, the training process was replicated using the four-class model. Figs. 3a and 3b show the training curves for the four-class model. Fig. 4 shows its confusion matrix. Table II compares F1-scores for shared classes across both models.

In terms of testing accuracy, the four-class model performed slightly better at 92.72%, whereas the five-class model scored

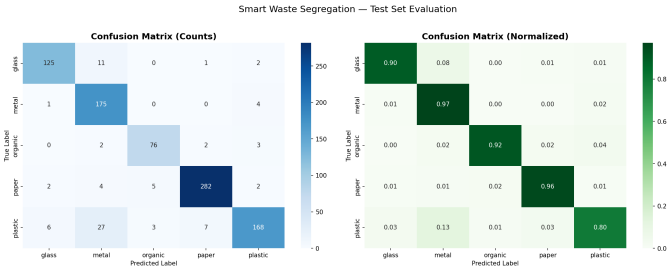
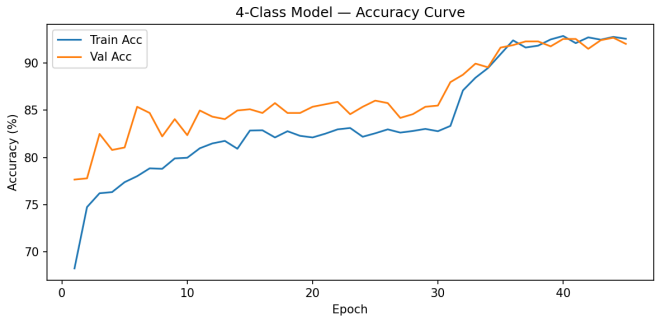
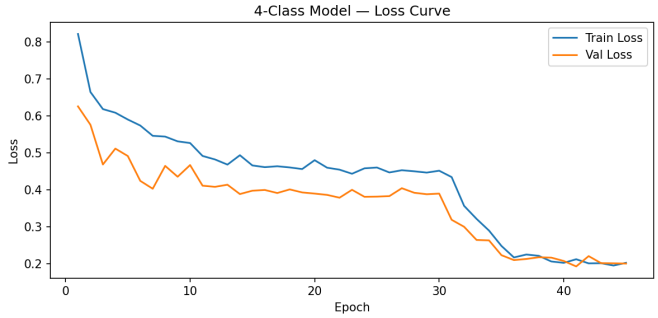


Fig. 2: Confusion matrix (counts and normalized) for the 5-class test set.



(a) 4-Class model training and validation accuracy over 45 epochs.



(b) 4-Class model training and validation loss over 45 epochs.

Fig. 3: 4-Class model training curves.

at 90.97%. The removal of glass had positive effects on plastic class F1 by 0.028 and organic by 0.023, which proved that glass causes cross-class confusion with plastic. Fig. 5 visualizes the per-class F1-score comparison between both models.

D. Real-Time Webcam Evaluation

The training model was deployed for testing in real-time using the laptop webcam under normal indoor lighting conditions without the use of GPU hardware. Five representative household objects were sequentially presented to the camera; those included the following objects: a printed paper document (paper); a glass bottle cap (glass); a hair dryer (metal); a plastic hot pack with red beads inside (plastic); an onion peel (organic). In all cases, the five objects were recognized correctly, with high confidence levels exceeding the

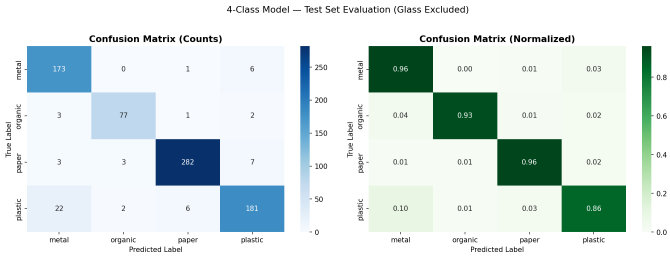


Fig. 4: Confusion matrix for the 4-class model (glass excluded).

TABLE II: Per-Class F1 Comparison, 4-Class vs. 5-Class Model

Class	4-Class F1	5-Class F1	Change
Metal	0.908	0.877	+0.031
Organic	0.933	0.910	+0.023
Paper	0.964	0.961	∼0.003
Plastic	0.889	0.862	+0.028

50% warning threshold. The predicted class, confidence bar and recommended waste disposal instruction were displayed accordingly in real-time, such as "Glass: Dispose in the Glass Recycling Bin" and "Organic: Dispose in the Compost / Green Waste Bin". Color feedback operated accordingly; all high-confidence predictions were highlighted by green color. Moreover, object removal resulted in correct prediction within one or two frames in all test cases. Fig. 6 shows the real-time webcam classification results for all five waste categories.

V. Discussion and Summary

This project proves that a simple transfer learning strategy based on MobileNetV2 can produce satisfactory five-class waste classification results in real-time deployment on consumer devices. The final network reaches 90.97% test accuracy and 0.905 macro F1 score, comparable to previous methods employing heavy networks on similar datasets.

The ablation experiment shows that there is a trade-off in the glass class. Glass removal increases the performance of plastic and organic classes by lowering visual confusion. However, the five-class model is still the proper architecture choice for practical applications because the network without glass cannot classify glass waste, thus failing to offer specific waste disposal advice for glass objects. This reduction of 1.75 percentage points can be adequately justified by the extra abilities.

The major limitation in this case would be the size of the training set. In both cases, only pictures captured under laboratory conditions of clean objects were used. In practice, the waste can be dirty, hidden, or mixed with other materials. Further studies can include datasets featuring different capture settings, test-time augmentation, or detection before classification when multiple waste objects are present in a scene.

In conclusion, the current project was successful in developing all the proposed components, which include a five-class CNN classifier, a web camera interface that shows disposal recom-

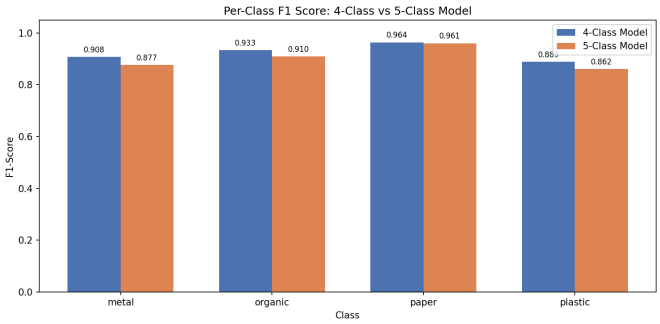
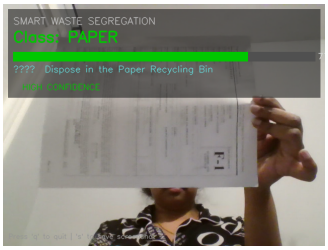
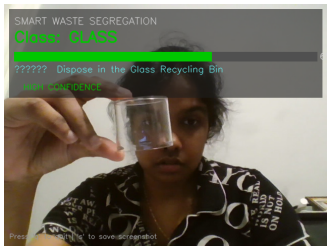


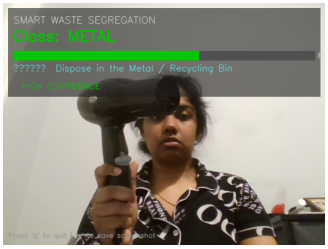
Fig. 5: Per-class F1-score comparison between the 4-class and 5-class models.



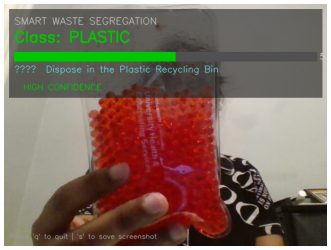
(a) Paper — document



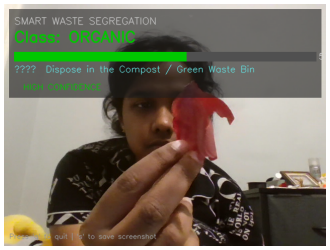
(b) Glass — container cap



(c) Metal — hair dryer



(d) Plastic — hot pack



(e) Organic — onion peel

Fig. 6: Real-time webcam classification results for all five waste categories. All predictions were correct with high confidence.

mendations and confidence levels, and a comparison of four- and five-class models. The project fulfills all design objectives and is useful in waste sorting applications.

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